IMPLEMENTATION / OPERATIONS / INCIDENT ANALYSIS

# AWS to Alibaba Cloud CEN

## *Complete IPsec VPN and BGP Technical Documentation*

Configuration history, annotated screenshots, log evidence, troubleshooting methodology, verified resolution, outstanding issues and operational validation.

Evidence period: 15-18 October 2026 | Report date: 20 October 2026

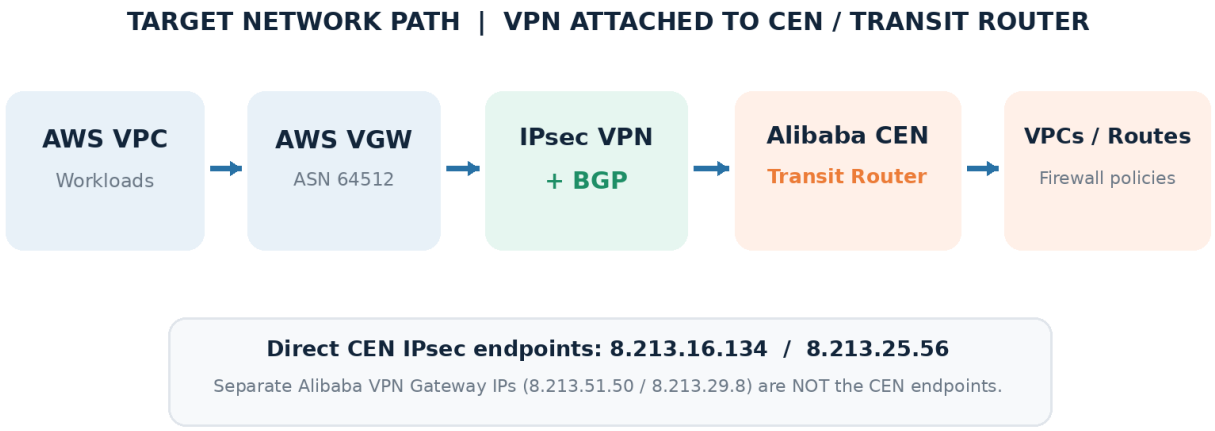



***Figure 1. Target topology: AWS VGW -> encrypted IPsec tunnel(s) -> Alibaba CEN / Transit Router. The separate Alibaba VPN Gateway is not the chosen endpoint.***

### Latest confirmed state

| VERIFIED SUCCESS | UNRESOLVED / UNTESTED |
|---|---|
| Alibaba CEN Tunnel 2: Phase 2 negotiation succeeded; BGP Normal. | Alibaba CEN Tunnel 1: Phase 1 negotiation failed; BGP abnormal. |
| AWS ipsec2 Tunnel 1: UP; 2 BGP routes displayed. | High-availability/failover tests, end-to-end workload connectivity and firewall transit not verified. |

Security note: this document intentionally excludes personal names and plaintext pre-shared keys. Screenshots are cropped to exclude credentials where practicable; exposed keys require rotation.

# Document guide

This is a detailed engineering case study, not a vendor-provided configuration export. The order follows the actual troubleshooting path and distinguishes verified evidence from hypotheses and future work.

| Section | Contents |
|---|---|
| 01 Executive assessment | Observed outcome, precise evidence limits and root-cause confidence |
| 02 System architecture | Network path, AWS VGW vs Alibaba VPN Gateway vs direct CEN attachment |
| 03 Component reference | Core networking terms and what each control does |
| 04 Resource inventory | AWS and Alibaba IDs, public endpoints, ASN and tunnel-inside CIDRs |
| 05 Implementation chronology | The full history: creation, dead ends, corrections and test results |
| 06 Issue analyses A-F | Endpoint confusion, missing ASN, wrong pairing, IKE/PSK, BGP, routing |
| 07 Working path evidence | Final screenshots showing IPsec and BGP health |
| 08 Outstanding path | Targeted troubleshooting plan for second CEN tunnel |
| 09 Validation playbook | Verification steps for security, routes, failover and operations |
| 10 Change controls | Safety, credential handling, rollback and monitoring |
| 11 Glossary & FAQ | Plain-English definitions and frequent misunderstandings |
| 12 Evidence and references | Figure source index, log provenance and official vendor references |

## Evidence classification

| Label | Meaning |
|---|---|
| OBSERVED | Displayed directly in a user-provided screenshot or log excerpt. |
| REASONED | A technical interpretation supported by observations but not independently proven. |
| RECOMMENDED | A proposed next action, not verified as implemented. |
| UNVERIFIED | No screenshot, current console query or network test was supplied. |

Important: a historical screenshot is a point-in-time observation. A configuration can change after it was captured; reported dates do not imply continued health on 20 October.

# 01 Executive assessment

## 1.1 Objective

Establish private-network reachability between AWS workloads and networks reached through Alibaba Cloud CEN by building encrypted public-internet Site-to-Site IPsec connectivity, routing with BGP, and an intended two-path design. The AWS side uses a Virtual Private Gateway (VGW). On the Alibaba side, the final chosen design attaches IPsec-VPN directly to a CEN Transit Router, rather than terminating on a separately provisioned Alibaba VPN Gateway.

## 1.2 Results captured on 18 October 2026

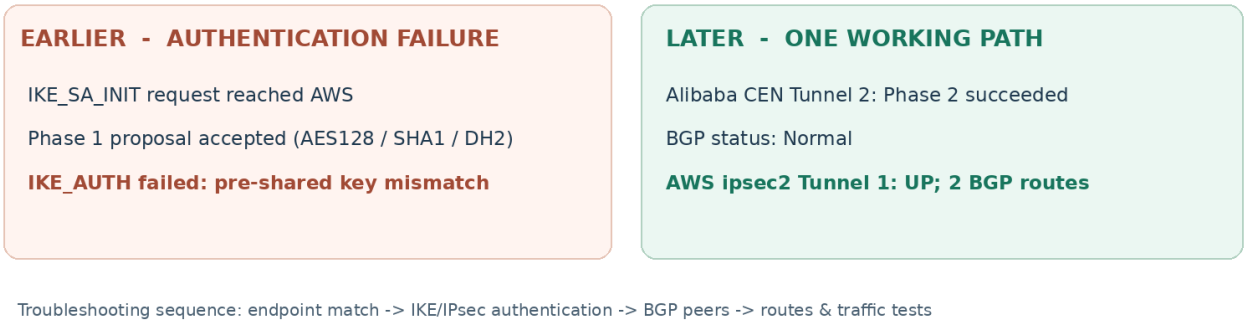



*Figure 2. Conceptual evidence progression. Left: earlier IKE authentication errors. Right: later confirmed success on one path. This is a schematic, not a console screenshot.*

| Check | CEN Tunnel 1 + AWS legacy | CEN Tunnel 2 + AWS ipsec2 |
|---|---|---|
| IKE/IPsec | Phase 1 failed (observed) | Phase 2 succeeded (observed) |
| BGP | Abnormal (observed) | Normal (observed) |
| AWS endpoint | 3.225.109.249 | 18.214.107.204 |
| AWS VPN resource | vpn-08155e0db7abb0798 | vpn-04caa84906b89e19d |
| Application traffic | Not validated | Not validated |

## 1.3 What was and was not resolved

- Confirmed: both clouds can exchange IKE packets; the earlier logs show proposal selection followed by IKE_AUTH authentication failure.
- Confirmed: on the later CEN Tunnel 2 / AWS ipsec2 Tunnel 1 pair, IPsec and BGP were operational in the final screenshots.
- Not confirmed: the previously observed PSK mismatch disappeared on CEN Tunnel 1; the final screenshot still shows that path down.
- Not confirmed: 2 BGP routes guarantee correct VPC route-table propagation, firewall policy, application reachability or high availability.

**Overall outcome: PARTIALLY OPERATIONAL - one cross-cloud encrypted and dynamically routed path; redundancy remains incomplete.**

# 02 System architecture and traffic flow

## 2.1 End-to-end design

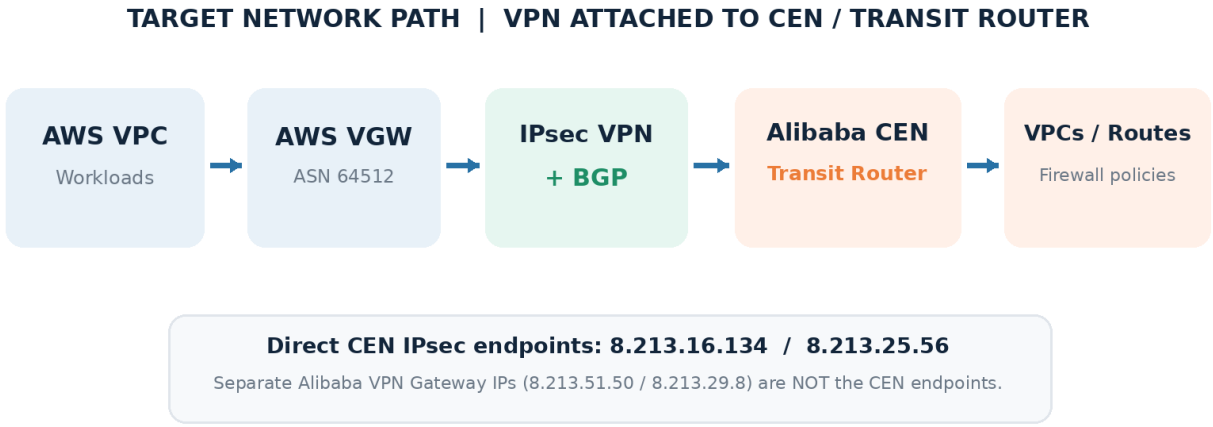



*Figure 3. Reconstructed logical topology; CEN route tables and network controls govern onward forwarding to VPC networks.*

1. An AWS workload sends traffic toward a nonlocal private CIDR; the effective subnet route table and the VGW association determine whether traffic is sent toward the VPN.
2. The AWS VGW and Alibaba CEN IPsec endpoint exchange IKE messages over their public addresses; authenticated IPsec Security Associations encrypt the selected data-plane traffic.
3. Inside the established encrypted tunnel, BGP peers exchange reachable prefixes over 169.254.x.x /30 addresses.
4. CEN route association, learning/propagation and policy determine how packets reach attached Alibaba VPCs. Firewall, security-group and ACL enforcement may still block traffic.
5. Return routes must exist in the reverse direction; otherwise connections can fail even while IPsec and BGP are both UP.

## 2.2 The two Alibaba connectivity products were easy to confuse

| Separate Alibaba VPN Gateway | Direct CEN IPsec attachment |
|---|---|
| Gateway public IPs: 8.213.51.50 and 8.213.29.8. | Final CEN connection public IPs: 8.213.16.134 and 8.213.25.56. |
| Represents IPsec service attached to Alibaba VPC-AWS-ABC. | Represents IPsec connection directly associated with Alibaba CEN / TR. |
| Was NOT the ultimately requested ingress path. | This is the chosen target architecture. |

The existence of both products does not imply an automatic relationship. A Customer Gateway configured with the public IP of one product will not correctly identify the other product.

# 03 Core networking concepts - practical explanations

| Term | Role in this project | What it is NOT |
|---|---|---|
| VPC | Private address space and subnets hosting resources. | Not an IPsec VPN tunnel. |
| VGW | AWS-managed VPN termination attached to one AWS VPC. | Not the external Alibaba peer. |
| AWS Customer Gateway | AWS configuration object describing the remote Alibaba public endpoint and its ASN. | Not a newly provisioned physical gateway. |
| Alibaba Customer Gateway | Alibaba object describing the remote AWS outside tunnel IP and ASN. | Not the Alibaba CEN Transit Router itself. |
| IPsec tunnel | Encrypted network-level channel between two public VPN endpoints. | Not a BGP session by itself. |
| IKEv2 | Negotiation protocol used to select cryptographic parameters and authenticate peers. | Not application-layer encryption. |
| PSK | Secret used to authenticate the two IKE peers. | Not interchangeable between unrelated tunnels unless configured identically. |
| BGP | Dynamic route exchange across the tunnel using a neighbor relationship. | Not responsible for the initial IKE handshake. |
| ASN | Identifier of a BGP autonomous system; required by the peers. | Not an IP address or route prefix. |
| Tunnel-inside /30 | Small link-local range used for the BGP peer addresses. | Not the workload CIDR being advertised. |
| CEN / TR | Alibaba inter-network routing fabric and transit-router attachment. | Not proof that firewall policies allow a workload flow. |

## 3.1 IKE and routing are different layers

IKE_SA_INIT is the first IKEv2 exchange; it negotiates parameters and key exchange. IKE_AUTH authenticates peers (including use of a PSK) and establishes authenticated IKE state. IPsec Child SAs then protect user traffic. BGP over the tunnel requires a usable encrypted path and correctly aligned neighbor addresses and ASNs. This layering is why a BGP Down message is often a downstream symptom when IKE Phase 1 is still failing.

## 3.2 Why public and private tunnel addresses must not be mixed up

Public outside addresses (for example 18.214.107.204) are used to carry IKE/IPsec traffic across the internet. Link-local inside addresses (for example 169.254.151.29) are used for BGP peering within the tunnel. VPC CIDRs (for example 10.0.0.0/16 in a separately observed Alibaba VPC) are application-routing prefixes, and must be validated before advertising them.

# 04 Verified resource inventory

## 4.1 AWS control-plane objects

| Resource | ID / observed value | Notes |
|---|---|---|
| AWS VGW | vgw-037425d907585c943 | State Available; VPC attachment state Attached |
| Amazon-side ASN | 64512 | Confirmed in AWS VGW Details screenshot |
| AWS VPC | vpc-0d3570ee490bf8dcc | CIDR not shown in evidence |
| VPN - legacy | vpn-08155e0db7abb0798 | Tunnel 1 outside 3.225.109.249 |
| VPN - ipsec2 | vpn-04caa84906b89e19d | Tunnel 1 outside 18.214.107.204 |
| AWS-side routing mode | Dynamic (BGP) | Observed on both VPN connections |

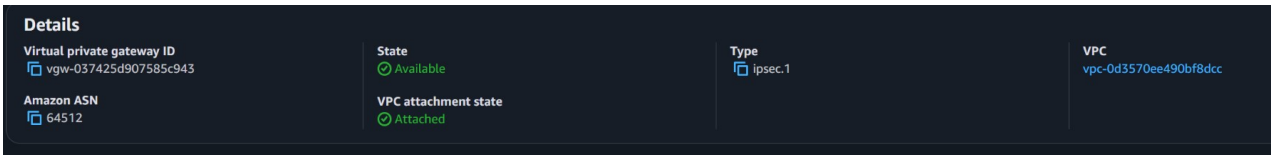



*Figure 4. AWS VGW: Amazon ASN 64512; resource Available and Attached.*

## 4.2 Alibaba control-plane objects

| Resource | ID / observed value | Notes |
|---|---|---|
| Active CEN IPsec | vco-l4vnrb880v8lim1me3fp3 | Associated with CEN; BGP enabled |
| Older CEN IPsec | vco-l4v8eqwgtzolwdbeudzxd | Earlier diagnostic path; different addresses |
| Alibaba BGP local ASN | 65000 | Matches the AWS Customer Gateway ASN setting |
| Separate Alibaba VPN Gateway | vpn-l4vtyi52sqo1gvq5wbaoe | Not the selected direct-CEN termination |
| Alibaba VPC-AWS-ABC | 10.0.0.0/16 | Separate VPC identified in screenshots |

# 04 Verified resource inventory (continued)

## 4.3 Definitive endpoint pairing at the latest checkpoint

| Key attribute | CEN Tunnel 1 / AWS legacy | CEN Tunnel 2 / AWS ipsec2 |
|---|---|---|
| Alibaba public outside IP | 8.213.16.134 | 8.213.25.56 |
| AWS Tunnel 1 outside IP | 3.225.109.249 | 18.214.107.204 |
| AWS Customer Gateway public IP | 8.213.16.134 | 8.213.25.56 |
| Alibaba AWS-facing Customer Gateway | 3.225.109.249 / ASN 64512 | 18.214.107.204 / ASN 64512 |
| Inside tunnel /30 | 169.254.63.84/30 | 169.254.151.28/30 |
| Alibaba local BGP address | 169.254.63.86 | 169.254.151.30 |
| AWS BGP neighbor IP (derived) | 169.254.63.85 | 169.254.151.29 |
| BGP ASNs: AWS / Alibaba | 64512 / 65000 | 64512 / 65000 |
| Last observed operational state | Phase 1 failed; BGP abnormal | IPsec Phase 2 success; BGP Normal |

The AWS neighbor addresses listed as "derived" are inferred from the AWS/Alibaba /30 information and should be verified against the latest AWS Download Configuration export before future changes. The tunnel CIDRs and Alibaba local addresses are explicitly visible in the final CEN screenshot.




*Figure 5. Latest legacy AWS VPN Customer Gateway IP: 8.213.16.134.*




*Figure 6. Latest ipsec2 AWS Customer Gateway IP: 8.213.25.56.*

# 05 Implementation history - chronological reconstruction

| Stage | Observed problem / action | What it taught us |
|---|---|---|
| 15 Oct - initial AWS | AWS Customer Gateways IP-SEC and IP1 pointed to 8.213.29.8 / 8.213.51.50; tunnels Down. | These addresses did correspond to an Alibaba gateway - but the wrong type for the chosen architecture. |
| 15 Oct - early IKE | Alibaba logs sent IKE traffic from 8.213.35.137 to AWS 3.225.109.249 with retries. | The observed source IP was not one of the initial AWS Customer Gateway addresses. |
| 18 Oct - CEN discovery | Found older CEN IPsec connection, which used 8.213.35.137 / 8.213.16.49. | The IKE logs belonged to a CEN-attached IPsec resource. |
| 18 Oct - BGP prerequisite | Alibaba displayed a direct error: Customer Gateway has no ASN and cannot support BGP. | The immediate blocker was missing remote BGP ASN on old Alibaba Customer Gateways. |
| 18 Oct - AWS expansion | Created second AWS VPN connection (ipsec2). | Two AWS VPN connections provided separate outside endpoints for two Alibaba CEN tunnels. |
| 18 Oct - new CEN | Created CEN-attached IPsec with BGP on, endpoints 8.213.16.134 / 8.213.25.56. | The new endpoint addresses had to be captured from the actual CEN resource. |
| 18 Oct - mapping corrected | Updated Customer Gateway associations on AWS and Alibaba to point to opposite peer addresses. | One mismatched endpoint pair can stop IKE before routing is relevant. |
| 18 Oct - IKE forensic | AWS logs: Phase 1 proposal accepted, but pre-shared key mismatch in IKE_AUTH. | At that moment, packet reachability and cryptographic proposal overlap were present; authentication still failed. |
| 18 Oct - BGP alignment | Updated Alibaba CEN /30 CIDRs and BGP IPs to match paired AWS Tunnel 1 networks. | BGP peer consistency is required after IPsec comes up. |
| 18 Oct - latest evidence | CEN Tunnel 2 Phase 2 succeeded and BGP Normal; AWS ipsec2 Tunnel 1 UP with 2 BGP routes. | One path was now working; the other remained unresolved. |

Sequence and labels are reconstructed from screenshots and messages; a precise audit trail of every Save/OK action is not available. Do not interpret the chronology as proof of exclusive causality for any individual setting.

# 06A Issue: incorrect Alibaba endpoint identity

## Problem

The first AWS Customer Gateway definitions targeted the two public IPs of the separate Alibaba VPN Gateway. The actual direct-CEN IPsec attachment eventually chosen for the project had different outside addresses. Despite both sets being valid-looking public IPs in Alibaba, they belonged to different logical resources.

## Symptoms and initial clues

- AWS displayed VPN Connection Available but both tunnels Down; Available described the provisioned AWS resource, not successful IKE/IPsec state.
- Earlier Alibaba IKE logs showed a source outside IP of 8.213.35.137, which did not match the initially configured AWS Customer Gateway addresses.




*Figure 7. Early AWS Customer Gateways created with public IPs 8.213.29.8 and 8.213.51.50; both objects Available.*




*Figure 8. Separate Alibaba VPN Gateway indeed owned 8.213.51.50 and 8.213.29.8, explaining why these initial mappings seemed plausible.*

## Investigation and resolution

After identifying that the target was CEN/Transit Router attachment rather than the standalone VPN Gateway, the AWS remote endpoint references were mapped to CEN attachment public addresses. The new active connection then showed CEN endpoints 8.213.16.134 and 8.213.25.56. Both AWS VPN connections were later shown pointing at the appropriate CEN addresses for their paired tunnels.

**Lessons learned: identify the target cloud resource and its public tunnel endpoint, not merely a public address within the same provider account.**

# 06B Issue: two CEN connections created during investigation

## Problem and evidence

An earlier CEN-attached IPsec resource (created 15 October) existed before the newer BGP-enabled CEN attachment (created 18 October). These are distinct resources. The earlier IPsec connection used CEN tunnel public IPs 8.213.35.137 and 8.213.16.49 and had negotiation failures. The later active connection used 8.213.16.134 and 8.213.25.56.

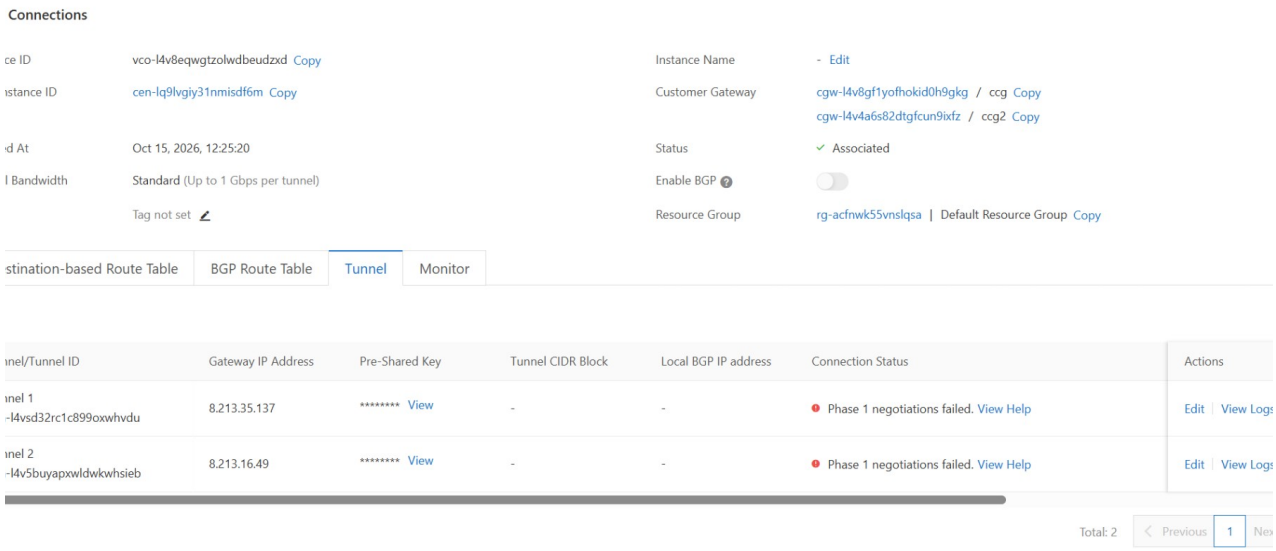



*Figure 9. Earlier CEN-attached connection (15 October). Phase 1 failures were associated with a different set of public tunnel IPs.*

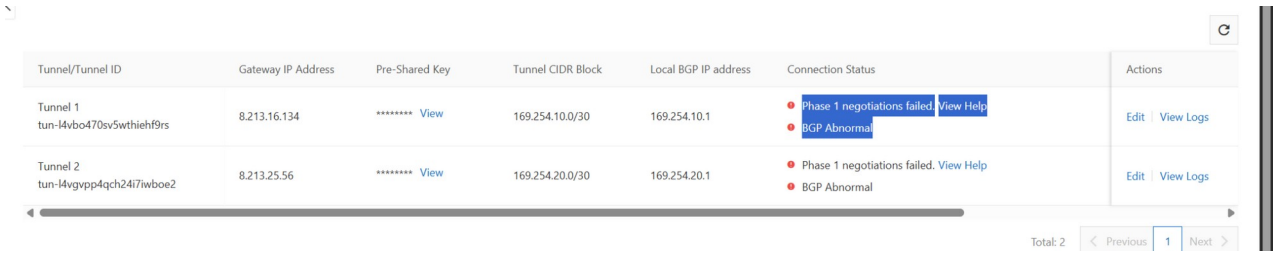



*Figure 10. New CEN IPsec connection (18 October): both tunnels initially reported Phase 1 failures and BGP Abnormal.*

## Why this complicated troubleshooting

- Logs and screenshots were drawn from different CEN resources at different times; mixing outside IP addresses caused false assumptions about which tunnel was initiating.
- Creating a new CEN IPsec attachment can generate new tunnel public IP addresses, invalidating any endpoint mapping copied from the earlier resource.
- Two CEN attachments may have separate costs and route/propagation effects. The data does not establish that the older attachment was later deleted or safely retired.

## Safe disposition

Maintain a resource inventory with lifecycle classification (active / superseded / candidate for removal) and verify route associations before deleting anything. Do not assume that creating a replacement automatically deactivates an older VPN attachment.

# 06C Issue: BGP unavailable due to missing ASN

## Exact blocker shown by console

> The customer gateway does not have an autonomous system number (ASN)
> and does not support BGP.

The initial Alibaba Customer Gateway objects (ccg and ccg2) used AWS public endpoints but showed a dash rather than a configured ASN. Attempting to enable BGP on the old CEN connection raised this explicit validation message.

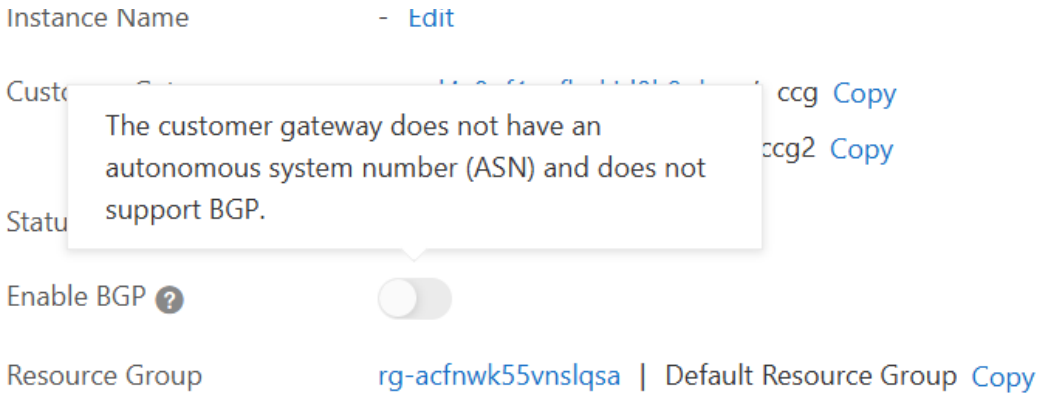



*Figure 11. The console identified the missing Customer Gateway ASN as the immediate BGP configuration blocker.*

## Diagnosis and corrective work

6. Confirmed Amazon-side ASN = 64512 using AWS -> VPC -> Virtual Private Gateways -> VGW details.
7. Created new Alibaba Customer Gateway objects for AWS public tunnel endpoints, with remote ASN 64512.
8. Configured Alibaba IPsec local ASN = 65000, corresponding to the ASN configured for Alibaba in AWS Customer Gateway objects.
9. Enabled BGP on the new CEN IPsec connection and then defined /30 and local BGP addresses per tunnel.

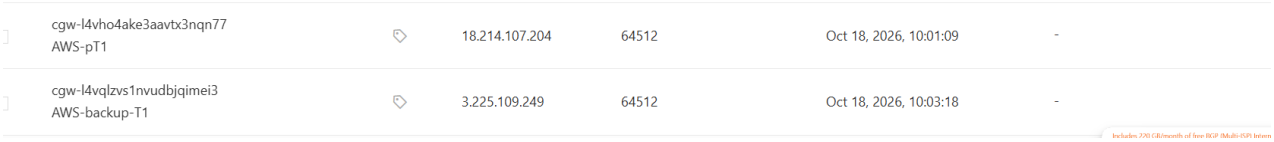



*Figure 12. New Alibaba Customer Gateways for 18.214.107.204 and 3.225.109.249, each with ASN 64512.*

ASN direction matters: AWS customer-gateway ASN is the Alibaba BGP ASN (65000), whereas Alibaba customer-gateway ASN is the Amazon-side VGW ASN (64512). These values are not meant to be identical.

# 06D Issue: AWS-to-Alibaba endpoint pair mismatch

## Definition of the error

Each Alibaba CEN tunnel must identify the AWS outside IP for the paired AWS tunnel; each AWS VPN connection must identify the Alibaba public endpoint used by its corresponding CEN tunnel. During configuration, the new AWS ipsec2 VPN was temporarily pointed at Alibaba CEN Tunnel 2 (8.213.25.56) even while logs from CEN Tunnel 1 were attempting the ipsec2 Tunnel 1 outside IP (18.214.107.204). The attempted pairings were inconsistent until both sides were remapped.

## Correct final mapping

| End | Alibaba public | AWS public | Paired resource |
|---|---|---|---|
| CEN Tunnel 1 | 8.213.16.134 | 3.225.109.249 | AWS legacy Tunnel 1 |
| CEN Tunnel 2 | 8.213.25.56 | 18.214.107.204 | AWS ipsec2 Tunnel 1 |

## Why this can break IKE

The outside public IP and IKE peer identity are used to determine which configured security association is being negotiated. A valid IKE packet reaching the wrong configured peer/remote identity might be rejected or be matched against incorrect configuration. The provided record demonstrates mapping confusion; the later CloudWatch log is stronger evidence of the contemporaneous authentication failure.

## How it was diagnosed

- Compared the Alibaba CEN "Gateway IP Address" values with AWS "Customer gateway address" fields.
- Compared Alibaba "Customer Gateway" selections and RemoteId with AWS "Outside IP address" on the specific tunnel (not just the VPN connection name).
- Checked AWS VPN Details after association changes, noting the old and new connections both target the expected Alibaba CEN addresses in their later screenshots.

**Verification criterion: verify FOUR values for every pair: Alibaba public outside IP, AWS CGW public IP, AWS outside tunnel IP, Alibaba remote AWS gateway/identity.**

# 06E Issue: IKE_SA_INIT then IKE_AUTH PSK mismatch

## What AWS actually logged

At 08:45:07 UTC on 18 October, AWS logged incoming IKEv2 traffic to both AWS Tunnel 1 outside endpoints. It selected a valid Phase 1 proposal and exchanged IKE_SA_INIT messages using UDP 500. A subsequent IKE_AUTH request arrived on UDP 4500, but AWS reported an explicit pre-shared key mismatch. This is a stronger diagnosis than a generic timeout or "Phase 1 failed" indicator.

```
08:45:07Z  received packet ... [UDP 500] ... IKE_SA_INIT
08:45:07Z  Selected proposals for Phase 1 SA:
        Encryption: AES128  Hash: SHA1  PRF: SHA1  DH Group(s): 2
08:45:07Z  received packet ... [UDP 4500] ... IKE_AUTH
08:45:07Z  AWS tunnel detected a pre-shared key mismatch
```

This is a shortened, non-secret excerpt preserving the diagnostic semantics of the provided CloudWatch events. The same authentication error was seen for both AWS Tunnel 1 pairings at the earlier checkpoint.

## Interpretation - which layers worked?

| Stage | Observed evidence | Diagnostic conclusion |
|---|---|---|
| Network initiation | Received IKE_SA_INIT UDP 500 | Basic ingress reachability to AWS endpoint existed at that time. |
| Proposal negotiation | Selected AES128 / SHA1 / DH 2 | At least one cryptographic Phase 1 proposal was mutually accepted. |
| Authentication | IKE_AUTH received on UDP 4500 | Handshake progressed beyond initial negotiation. |
| IKE peer authentication | Explicit pre-shared key mismatch | Authentication configuration was inconsistent at that checkpoint. |
| IPsec Child SA / BGP | Down at same checkpoint | Subsequent layers could not be assumed healthy. |

## Resolution and evidence limits

Shared secret values were subsequently aligned between counterpart tunnels, alongside other IKE and BGP edits. The final status confirms authenticated IPsec connectivity on CEN Tunnel 2; it does not establish which one change alone resolved that tunnel, and it does not confirm the same resolution for CEN Tunnel 1. AWS official documentation associates this specific message with inconsistent PSK configuration between IKE peers [V2].

# 06F Issue: IKE identities, parameters and rekey behavior

## Configuration comparison

| Parameter | Alibaba setting seen | AWS allowed/observed | Implication |
|---|---|---|---|
| IKE version | IKEv2 | IKEv1 and IKEv2 permitted | Compatible; prefer IKEv2. |
| Encryption | AES128 | AES128 among proposals | Compatible for the logged IKE selection. |
| Integrity / PRF | SHA1 | SHA1 among proposals | Compatible for the logged IKE selection. |
| DH group | group2 | 2 among allowed groups | Compatible but cryptographically dated. |
| Phase 1 lifetime | 86400 seconds in Alibaba initial view | 28800 seconds in AWS | Review harmonization; not established as PSK error root cause. |
| Phase 2 lifetime | Not fully evidenced in same view | 3600 seconds shown in AWS | Reconcile during planned hardening. |
| LocalId / RemoteId | A stale RemoteId was seen during mapping change | AWS IKE peer identity settings not fully exported | Confirm identities after endpoint remapping. |

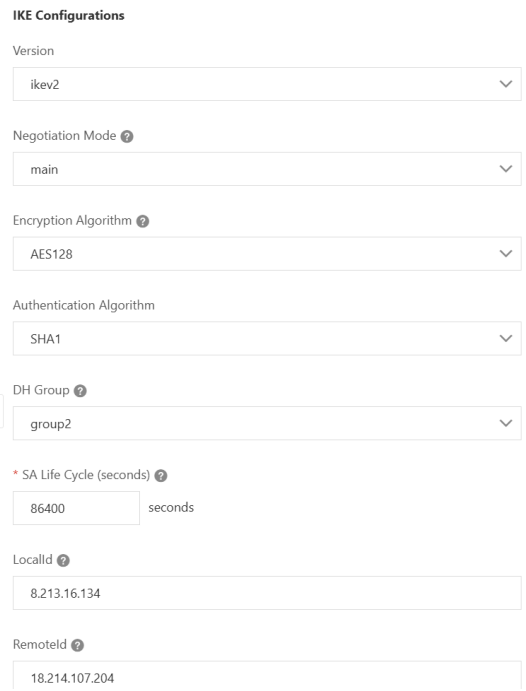



*Figure 13. Alibaba IKE configuration during the early editing period: IKEv2, AES128, SHA1, DH group 2, and Phase 1 lifetime 86400 s.*

## Why IDs matter

LocalId identifies the Alibaba peer during IKE authentication; RemoteId identifies the expected AWS peer. When the paired AWS outside tunnel IP changes, these identifiers must be rechecked rather than assumed to update automatically with the selected Customer Gateway. Use the identities specified by the authoritative current AWS VPN configuration. Do not simply copy an outside IP blindly when identity customization is in use.

## Security improvement - after service stabilization

AES128/SHA1/DH2 was negotiated successfully in a logged Phase 1 proposal, so it is not necessary to change all algorithms to fix the current outage. However, SHA1 and DH group 2 are legacy choices; a planned change should move to an agreed strong IKEv2 cipher suite (for example SHA2-256 with DH group 14 or stronger), first confirming support on both cloud endpoints. Avoid hardening a working tunnel during unfinished incident isolation.

## Identity and proposal cross-check worksheet

| Peer association | What to cross-check in the current saved configuration |
|---|---|
| CEN Tunnel 1 / legacy AWS | Alibaba LocalId matches its own 8.213.16.134 peer identity; RemoteId corresponds to the AWS 3.225.109.249 peer configuration; PSK matches that exact AWS tunnel. |
| CEN Tunnel 2 / AWS ipsec2 | Alibaba LocalId matches 8.213.25.56; RemoteId corresponds to AWS 18.214.107.204; the separately paired PSK matches. |
| Cryptographic settings | An overlapping IKEv2 proposal exists; AWS earlier accepted AES128/SHA1/DH2, but current policy and Phase 2 PFS still require validation. |
| Lifecycle / rekey | Review Phase 1/Phase 2 lifetimes, DPD timeout, startup action and any configured NAT traversal behavior, especially after a fresh change. |

The correct technical test is to compare the saved peer parameters and examine NEW log entries after a change, not to infer success from values visible on an unsaved edit form. The already-operating CEN Tunnel 2 provides a useful control case, but copying its current PSK onto unrelated peers should not replace deliberate key management.

# 07 BGP tunnel network alignment

## The /30 issue

When the new Alibaba CEN IPsec was first provisioned, its BGP inside subnets were 169.254.10.0/30 and 169.254.20.0/30 with Alibaba local BGP IPs 169.254.10.1 and 169.254.20.1. AWS Tunnel 1 on the two paired AWS VPNs had independently assigned 169.254.63.84/30 and 169.254.151.28/30. A BGP session cannot function with peers configured for unrelated link networks.

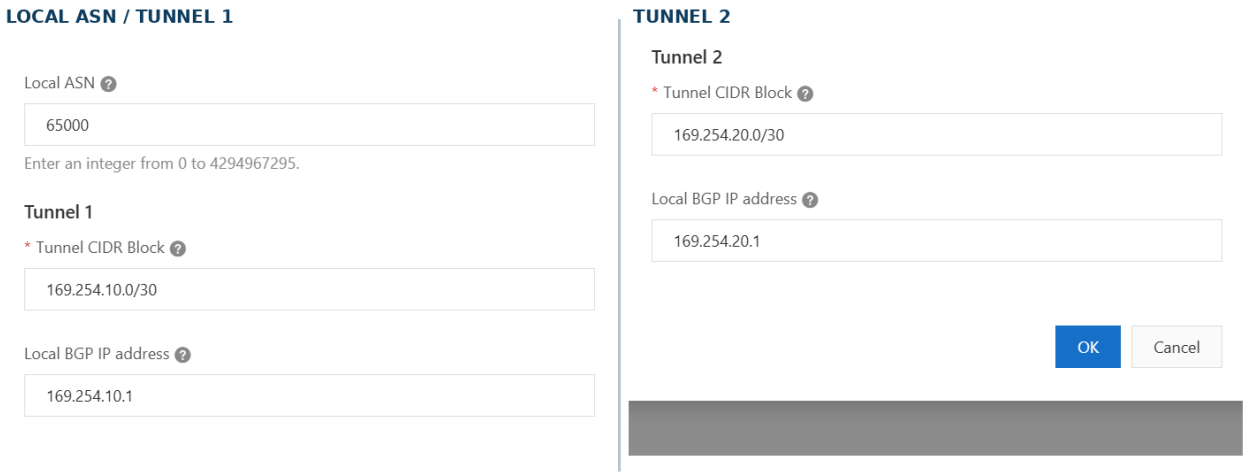



*Figure 14. Early BGP dialog with mismatched Alibaba tunnel-inside /30 prefixes before alignment.*

## Applied values visible in final Alibaba screenshot

| Parameter | CEN Tunnel 1 | CEN Tunnel 2 |
|---|---|---|
| Tunnel CIDR | 169.254.63.84/30 | 169.254.151.28/30 |
| Alibaba local BGP address | 169.254.63.86 | 169.254.151.30 |
| AWS peer (derived, verify in export) | 169.254.63.85 | 169.254.151.29 |
| Alibaba local ASN | 65000 | 65000 |
| AWS ASN | 64512 | 64512 |

## Why use /30 link-local addresses?

Each /30 allocates four addresses: a network address, two usable host addresses and a broadcast address. For 169.254.151.28/30 the usable pair is .29 and .30. These addresses identify the BGP speakers inside the tunnel; they are not public IPs and should not be advertised as workload networks.

The latest Alibaba CEN status picture verifies the new CIDRs and its own local BGP IPs. The final working BGP session on Tunnel 2 is evidence that the route exchange is valid for that pair. For Tunnel 1, a correct /30 still cannot overcome its separate IKE Phase 1 failure.

# 08 BGP routing, CEN policy and network-layer validation

## 8.1 BGP UP versus application reachable

An AWS console reporting "2 BGP ROUTES" tells us that the AWS VPN control plane is receiving or associating BGP route information at that moment. It does NOT establish that every required private subnet route is installed in the effective AWS VPC subnet route table, or that the intended destination is propagated through CEN to an Alibaba VPC. It also does not prove an allowed return path.

## 8.2 Route tables involved

| Layer | Verification question |
|---|---|
| AWS VPC subnet route table | Are destination Alibaba CIDRs directed toward the attached VGW or applicable route propagation? |
| AWS VGW / BGP | Are the expected remote prefixes learned, accepted, and preferred? |
| Alibaba IPsec attachment route table | Are AWS prefixes learned and remote Alibaba prefixes advertised with correct scope? |
| Alibaba CEN Transit Router route table | Are the IPsec and VPC attachments associated with the intended TR route tables? |
| CEN propagation / export policy | Are only approved VPC CIDRs propagated? Is the forwarding chain valid in both directions? |
| Security controls | Do Cloud Firewall, Security Groups and Network ACLs permit the specific protocol/port? |
| Application | Can a selected host reach a selected remote service and receive a response? |

## 8.3 Advanced create-time options

The Alibaba interface exposed options for automatically advertising routes to the VPN, associating the attachment with the Transit Router system route table, and propagating routes. These should be reviewed as change-controlled routing-policy decisions, especially in a shared company CEN. Their checked state in a creation screenshot was not independently reverified after deployment.

Alibaba documentation describes dual-tunnel active-active ECMP behavior for CEN-attached IPsec when both routes are available [V5]. AWS may also carry additional unused provider-managed tunnels within each Site-to-Site VPN connection; this design paired only Tunnel 1 from each of two AWS connections. Test failover on the actual route path rather than relying on individual AWS "not using both tunnels" banners.

# 09 Final verification - Alibaba CEN (captured)

The final Alibaba screenshot is the most useful high-level end-state evidence because it shows both CEN tunnels side by side and their BGP settings, preventing an optimistic interpretation based only on the working tunnel.

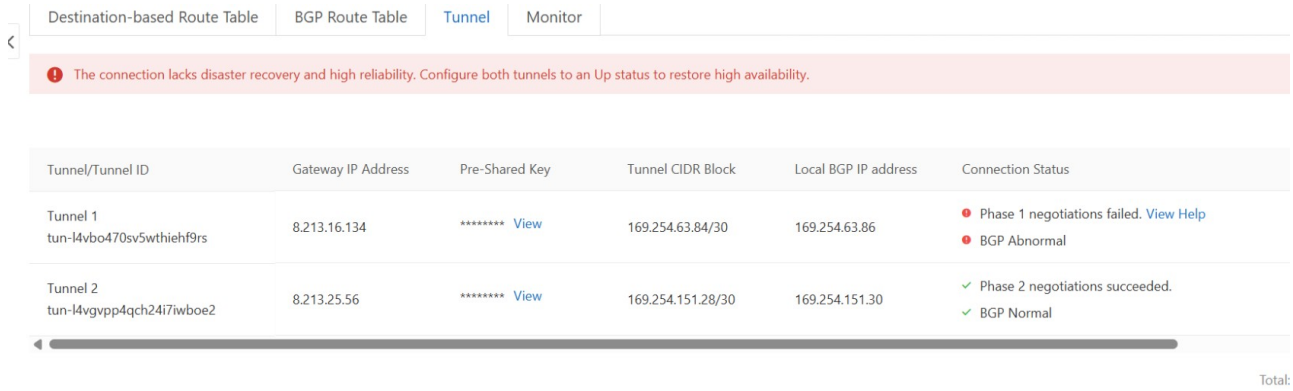



*Figure 15. Final CEN status: Tunnel 1 Phase 1 failed / BGP Abnormal; Tunnel 2 Phase 2 succeeded / BGP Normal.*

## What this proves

- CEN Tunnel 2 reached the negotiated IPsec data-protection state (reported Phase 2 succeeded).
- The BGP neighbor on CEN Tunnel 2 reached an acceptable operational state (BGP Normal).
- CEN Tunnel 1 was NOT healthy: IKE Phase 1 and BGP remained failed/abnormal.
- The CEN service itself reported insufficient high-availability/disaster-recovery coverage until both tunnels are Up.

## Evidence boundary

The screenshot shows status only. It does not provide per-prefix export or acceptance lists, active packet counters, throughput history, or workload traffic testing. Consequently, "working cross-cloud tunnel and BGP" is a supported conclusion; "production-ready intercloud service" is not yet supported.

# 09 Final verification - AWS (captured)

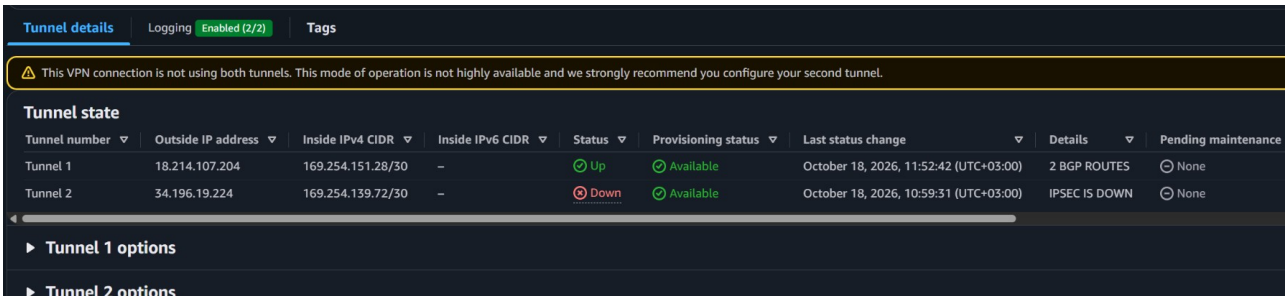



*Figure 16. Final AWS ipsec2 state: Tunnel 1 Up with 2 BGP Routes; AWS Tunnel 2 Down.*

## Cross-provider verification

| Evidence | Interpretation |
|---|---|
| Alibaba CEN Tunnel 2: Phase 2 succeeded | Alibaba recognized an established IPsec tunnel. |
| Alibaba CEN Tunnel 2: BGP Normal | Alibaba recognized an operational dynamic-routing peer. |
| AWS ipsec2 Tunnel 1: Up | AWS recognized the paired tunnel as operational. |
| AWS ipsec2 Tunnel 1: 2 BGP ROUTES | AWS exposed learned/associated route data for the established tunnel. |
| Other CEN tunnel still failed | This is not full dual-path high availability. |

## How to interpret AWS warning correctly

AWS warns when a Site-to-Site VPN is not using both of its own built-in tunnels. In the captured CEN design, Alibaba Tunnel 1 was paired with the first tunnel of the legacy AWS VPN and Alibaba Tunnel 2 with the first tunnel of ipsec2. Therefore this AWS banner is not, by itself, sufficient to evaluate redundancy of the CEN dual-tunnel service. The Alibaba CEN warning and real failover tests are decisive: the CEN had only one healthy path in the latest supplied view.

# 10 Remaining problem: CEN Tunnel 1 fails IKE Phase 1

## Path under investigation

| Field | Expected / last seen |
|---|---|
| Alibaba CEN outside IP | 8.213.16.134 |
| AWS legacy VPN Tunnel 1 outside IP | 3.225.109.249 |
| AWS Customer Gateway association | 8.213.16.134 (latest screenshot) |
| Alibaba Customer Gateway | AWS 3.225.109.249; remote ASN 64512 |
| Inside network / local BGP IP | 169.254.63.84/30 / Alibaba 169.254.63.86 |
| Current status in last screenshot | Phase 1 failed; BGP Abnormal |

## Hypothesis queue ranked by evidence

| Hypothesis | Evidence priority / way to test |
|---|---|
| PSK mismatch remains on legacy pair | HIGH: Earlier AWS log explicitly recorded this. Recheck fresh, post-change IKE_AUTH log. |
| LocalId / RemoteId not aligned | HIGH-MEDIUM: stale remote identity seen during edit; compare current AWS peer identity and Alibaba IDs. |
| Unsaved / unapplied changes | MEDIUM: several configuration screens were shown before confirmation; verify saved values, not edit drafts. |
| IKE proposal mismatch | LOWER: AWS previously accepted AES128/SHA1/DH2; only re-evaluate if new logs now reject proposals. |
| Transport block on UDP 500/4500 | LOWER at earlier checkpoint: AWS received both UDP 500 and 4500; re-evaluate only if latest logs are silent. |
| BGP or inside CIDR causing IKE failure | NOT primary: BGP is downstream of IKE Phase 1; inside /30 mismatch is a separate issue. |

A single line "Phase 1 negotiations failed" does not isolate a cause. The timestamped AWS log for the legacy Tunnel 1 after its last configuration save is the best next diagnostic artifact.

# 11 Repeatable troubleshooting runbook

## Runbook 1 - preserve the healthy tunnel first

10. Record the currently Up path and the latest CEN/AWS timestamps; do not modify AWS ipsec2 Tunnel 1 or Alibaba CEN Tunnel 2 during legacy-path investigation.
11. Confirm a maintenance window and change owner for any setting that causes an IKE rekey or tunnel replacement; capture the pre-change configuration safely.
12. Do not use screenshots that expose Pre-Shared Keys. Rotate secrets previously shown in chat or tickets.

## Runbook 2 - investigate IKE on the failed pair

13. Alibaba: IPsec Connections -> active CEN connection -> Tunnel -> Tunnel 1 -> View Logs. Capture the most recent attempts and exact timestamp/timezone.
14. AWS: VPC -> Site-to-Site VPN Connections -> legacy VPN -> its Tunnel 1 -> CloudWatch tunnel activity logs. Filter to the same time window.
15. Confirm source/destination outside IP pairs and UDP 500/4500; if no request reaches AWS, inspect mapping/connectivity, but do not assume firewall failure from a generic Down badge.
16. Find IKE_SA_INIT proposal-selection and IKE_AUTH outcome. A log of "pre-shared key mismatch" justifies peer PSK and identity validation; "No Proposal Match Found" shifts focus to supported encryption/integrity/DH parameters.
17. Coordinate a new unique secret for this pair using an approved secret manager/change process, apply it to both peers, and inspect fresh logs after propagation. Do not paste the secret into this document or chat.
18. After Phase 1 succeeds, verify IPsec Phase 2 success and BGP Normal. If Phase 2 fails, review Child-SA, selectors, PFS, traffic selectors, lifetime and peer identity based on the new error.

## Runbook 3 - confirm BGP

19. Compare AWS downloaded current configuration with Alibaba BGP dialog: 169.254.63.84/30, Alibaba 169.254.63.86, correct AWS peer, local ASN 65000, remote ASN 64512.
20. Inspect the BGP Route Table and AWS tunnel details. A valid BGP session must have the intended prefix set, not only an Up icon.

Vendor references for these procedures: [V2], [V3], [V5], [V8].

# 12 Acceptance testing - make the network service real

## 12.1 Functional test checklist

| Priority | Test | Pass condition |
|---|---|---|
| P0 | Both planned CEN tunnels healthy | Both: Phase 2 negotiation succeeded, BGP Normal. |
| P0 | AWS peer state | Each intended paired AWS Tunnel 1 shows UP. |
| P0 | BGP route review | Prefixes on both ends match the approved network list; no accidental default route. |
| P1 | AWS effective VPC routing | Application source subnet routes point toward the correct gateway. |
| P1 | CEN route association/propagation | Destination and return CIDRs are in the intended TR and attachment route tables. |
| P1 | Firewall policy | Only approved sources, destinations and ports are permitted. |
| P1 | Application validation | Bidirectional service-specific tests work using private addresses. |
| P1 | Return path | AWS and Alibaba respond on valid paths; no asymmetric route problem. |
| P2 | Failover / recovery | An approved controlled tunnel outage recovers traffic without violating the defined RTO. |
| P2 | Monitoring | Alarms cover tunnel Down, BGP Down, packet/byte activity and error logs. |
| P2 | Credential hygiene | All previously exposed PSKs rotated; unique per intentionally paired IKE relationship. |

## 12.2 Suggested test record fields

For each test, record timestamp (with timezone), source host/subnet, destination service/IP/port, cloud region, effective route, security policy, command/tool, result, latency/loss if relevant, and linked evidence. Test only approved hosts; do not assume permitted ICMP when the production protocol is TCP.

## 12.3 Failover test boundary

Failover should only be attempted when both planned CEN tunnels are healthy and the routing team has approved the exact blast radius. Check whether both routes are active/ECMP in the actual attached TR route table; otherwise active/standby behavior may differ. Validate business traffic before, during and after a controlled tunnel interruption.

# 13 Safe operational change and security practice

## 13.1 Change checklist

- Scope: identify the exact AWS VPN ID, AWS outside tunnel IP, Alibaba CEN connection ID and Alibaba tunnel ID. IDs prevent mistaken edits between two VPN connections.
- Record the before/after state of IPs, ASNs, /30, IDs, PSK metadata (never value), IKE version, algorithms and route-table attachment relationships.
- Prefer one change per diagnostic iteration. Changing peer, secret, algorithms and BGP CIDR simultaneously makes root-cause proof difficult.
- Modifying AWS tunnel options interrupts that tunnel for potentially several minutes; changes to VPN Customer Gateway association also require coordinated downtime planning.
- Preserve the healthy CEN Tunnel 2 / AWS ipsec2 Tunnel 1 while repairing the other path.
- Verify AWS and Alibaba log timestamps in UTC and local time before correlating events.

## 13.2 Credential handling

Several PSKs were inadvertently exposed in screenshots during troubleshooting, so treat them as compromised. Do not forward or reuse them. Create fresh high-entropy PSKs using an approved secret workflow, use a unique one per intended tunnel pair, and rotate them in a controlled maintenance window. The document includes no plaintext secret, and cropped screenshots suppress these fields where practical.

## 13.3 Logging and audit

Keep CloudWatch tunnel activity and BGP logs in a dedicated log group with retention and access controls. AWS logs explicitly distinguish IKE_SA_INIT, accepted cryptographic proposals, IKE_AUTH, PSK mismatch and BGP changes. On Alibaba, keep logs from each CEN tunnel together with tunnel ID, outside IP and timestamp. Avoid mixing logs from the older and newer CEN attachments.

## 13.4 Clean up obsolete resources only after validation

The record contains an older CEN-attached IPsec resource and a separate Alibaba VPN Gateway in addition to the newer direct CEN attachment. Their future disposition was not verified. Inventory route and cost dependencies before proposing any deletion; do not remove a resource solely because it did not appear in the final successful screenshot.

# 14 Frequently asked engineering questions

### Why does AWS say "Available" when Tunnel state is Down?

Available refers to the provisioned resource, whereas Tunnel state refers to operational IKE/IPsec establishment.

### Why can BGP say Down when IKE is failing?

A BGP TCP session depends on the tunnel-inside route; failure in IKE/IPsec prevents the required protected path.

### Can PSKs be the same on both tunnels?

Yes, technically, if each corresponding peer has exactly the same secret; however, separate unique PSKs are preferable for isolation and security.

### Does a working BGP session mean applications can communicate?

No. Prefix advertisements may be filtered or route tables, firewall rules, ACLs, security groups or return routes may still be incorrect.

### Why are there two VPN connections in AWS if each already has two AWS tunnels?

This topology used Tunnel 1 from separate AWS VPN connections to match the two distinct public outside endpoints of one Alibaba CEN dual-tunnel attachment.

### Does the AWS unused-tunnel banner prove there is no HA?

Not by itself. It describes an individual AWS VPN connection. The CEN status and tested end-to-end failover determine redundancy of this paired topology.

### Are SHA1/DH2 the confirmed cause of the latest outage?

No. AWS accepted those parameters at the earlier checkpoint, then specifically reported a PSK mismatch. Cryptographic modernization is a separate recommended improvement.

### Could a Phase 1 lifetime difference cause negotiation trouble?

It is a configuration discrepancy worth reconciling; the record does not prove it caused the logged authentication failure.

### Is BGP unsupported in the Riyadh region?

A vendor documentation list did not list Riyadh at the time of review, but actual captured CEN Tunnel 2 reported BGP Normal. The observed configured service worked on that path, and any regional policy question should be confirmed with the relevant provider account team.

### What is the immediate next action?

Pull fresh, post-change AWS IKE logs for the legacy VPN Tunnel 1 and correlate with CEN Tunnel 1 before making additional changes.

# 15 Figure and evidence index

| Fig. | Source / why it matters |
|---|---|
| 1, 3 | Reconstructed target architecture. A conceptual figure only. |
| 2 | Conceptual progression showing earlier authentication failure and later one-path success. |
| 4 | AWS VGW / Amazon-side ASN 64512. |
| 5-6 | Final AWS Customer Gateway assignments for the two VPN connections. |
| 7-8 | Initially used AWS Customer Gateway public IPs and separate Alibaba VPN Gateway IPs. |
| 9 | Earlier CEN IPsec attachment with different public endpoints. |
| 10 | New CEN connection both tunnels Phase 1 failed prior to changes. |
| 11 | Alibaba explicit BGP missing ASN error. |
| 12 | ASN-enabled Alibaba Customer Gateway objects for AWS endpoints. |
| 13 | Alibaba earlier IKE configuration and legacy cryptographic parameters. |
| 14 | Initial BGP /30 configuration before re-alignment. |
| 15-16 | Final paired Alibaba and AWS status: one working tunnel, one not yet restored. |

## 15.1 Data lineage and sanitization

Visual evidence was sourced from screenshots supplied during the troubleshooting conversation. Crops focus on resource status, relevant IP fields, route/BGP settings and error messages. Plaintext secrets and personal names were deliberately not included in the document. Shortened IKE log excerpts exclude opaque payload dumps and secret values; they retain the verified diagnostic statements. All IP addresses and resource IDs included are technical operational identifiers and should be treated as internal engineering documentation.

## 15.2 Assumptions and uncertainties

- This document does not assert a present-time live tunnel state, because there was no new API-based check on the date the report was assembled.
- AWS downloaded configuration files were not supplied; inferred AWS BGP peer addresses should be confirmed before future edits.
- No exhaustive traffic tests, CEN route-table export, firewall rule set, or cost audit was provided.
- The specific set of changes responsible for bringing CEN Tunnel 2 up was not isolated in a controlled one-variable-at-a-time test.

# 16 Official technical references

The sources below provide general engineering behavior and vendor instructions. They are not substitutes for the project-specific screenshots and timestamps used to establish what occurred.

**[V1] AWS Site-to-Site VPN concepts.** https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

**[V2] AWS Site-to-Site VPN logs and diagnostic messages.**
https://docs.aws.amazon.com/vpn/latest/s2svpn/monitoring-logs.html

**[V3] AWS VPN tunnel option modifications.** https://docs.aws.amazon.com/vpn/latest/s2svpn/modify-vpn-tunnel-options.html

**[V4] AWS Site-to-Site VPN routing.** https://docs.aws.amazon.com/vpn/latest/s2svpn/VPNRoutingTypes.html

**[V5] Alibaba IPsec-VPN attachment to Transit Router.** https://www.alibabacloud.com/help/en/vpn/sub-product-ipsec-vpn/user-guide/create-and-manage-ipsec-vpn-connections-dual-tunnel

**[V6] Alibaba IPsec-VPN routes through Transit Router.** https://www.alibabacloud.com/help/en/vpn/sub-product-ipsec-vpn/user-guide/configure-a-route-for-an-ipsec-vpn-connection

**[V7] Alibaba CEN VPN attachment workflow.** https://www.alibabacloud.com/help/en/cen/user-guide/attach-an-ipsec-vpn-connection-to-a-transit-router

**[V8] AWS IKE troubleshooting.** https://repost.aws/knowledge-center/vpn-ike-connection-failures

## Final engineering conclusion

The case progressed from two Down AWS VPN tunnels and uncertainty about the correct Alibaba endpoint to a demonstrably established IPsec and BGP path between Alibaba CEN Tunnel 2 and AWS ipsec2 Tunnel 1. Key blockers encountered were confusion between a standalone Alibaba VPN Gateway and direct CEN IPsec attachment, Customer Gateways lacking a BGP ASN, public endpoint pairing inconsistencies, misaligned inside BGP /30s, and a logged IKE_AUTH pre-shared key mismatch. The remaining CEN Tunnel 1 still reported Phase 1 failure in the last evidence. Production acceptance requires repairing that second path, rotating exposed keys, reviewing routes and security controls, testing actual private-workload connectivity, and validating failover.

END OF TECHNICAL DOCUMENTATION